

Trần Hoàng Khải

Back-end Developer

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CAREER OBJECTIVE

I recently graduated with a degree in Information Systems from VNUHCM - University of Science and am seeking full-time opportunities as a **Back-end Developer**. I have hands-on experience building back-end systems using **C#**, **ASP.NET Core**, **FastAPI**, and **Entity Framework Core** for .NET and **SQLAlchemy/SQLModel** for FastAPI, with practical experience in HRM systems and e-commerce platforms.

My current goal is to build a long-term commitment with the company in order to understand the work environment and gain valuable experience. I also aim to expand my knowledge about career development paths and future opportunities for personal growth in the IT field

EDUCATION

VNUHCM - UNIVERSITY OF SCIENCE (HCMUS) OCT. 2021 - Present

- BACHELOR OF INFORMATION TECHNOLOGY
- Major: Information Systems

PROJECT EXPERIENCE

Project: EC-AirBnB Auction 6/2025 - 9/2025
size : 4 people

Description:

- Full-stack Airbnb Clone Platform with clean architecture built using Next.js 15 frontend, FastAPI Python backend, and an advanced data crawling system. Features include real-time messaging via Pusher, PostgreSQL database with SQLAlchemy ORM, Redis caching, and RabbitMQ streaming for scalable property management, user authentication, and booking systems across Traveller, Host, and Admin interfaces..
- View project:
  - BE: https://github.com/vongagh094/ecommerce_be
  - FE: https://github.com/vongagh094/ecommerce_fe

Responsibilities:

- My task was to develop the real-time bidding feature. I used RabbitMQ to publish and subscribe to bids, combined with Redis cache to boost performance in storing and querying bid data.
- I established the clean architecture pattern, configured Docker for database containerization, and implemented dependency injection for this academic project.

Performances:

- **Redis Caching**: Reduced bid query time from 1000ms to <1ms, achieving 99.9% cache hit rate and supporting 1000+ concurrent users
- RabbitMQ Streaming: Implemented asynchronous processing, handling 10,000+ bids/second with <50ms latency and zero message loss

Technologies: FastAPI (Python) (Backend), SQLAlchemy (ORM), PostgreSQL (Database), Redis (Cache), RabbitMQ (Message Queue), Docker, Next.js - TypeScript (Frontend)

Project: Hospital management system 8/2025 - 9/2025
size: 4 people

Description:

- Developed a microservices-based hospital management system with FastAPI and PostgreSQL handling 100+ requests/second during peak hours. Achieved sub-1-second API response times through optimized database queries, Redis caching, and Docker containerization with RabbitMQ message queuing.
- View project: https://github.com/THKhai/UDPT-Hospital-Management-System-BE

Responsibilities:

- Developed a comprehensive JWT authentication system - Implemented user login, registration, and session management with automatic token expiration, fully integrated with the user interface
- Built user management APIs - Created RESTful endpoints for patient profile management, including displaying user information, updating profile data, and serving the dashboard/home page content

Technologies: Python (FastAPI, SQLAlchemy), HTML/CSS/JavaScript (Bootstrap 5), PostgreSQL, MongoDB, Docker, JWT Authentication, Rabbitmq

TECHNICAL SKILLS

- **Programming Languages**: C#, C++, Python
- **Back-end Development**:
  - ASP.NET Core, .NET Framework, RESTful APIs, Entity Framework Core
  - FastAPI, SQLAlchemy/SQLModel
- **Database Systems**: Microsoft SQL Server, MongoDB, Redis, Cassandra, Neo4j, Oracle, Postgres
- **Tools & Technologies**: Docker, Dependency Injection, JWT, ASP.NET Identity, Swagger, CORS
- **Front-end Development**: WinForms, React.js
- **Data Skills**: ETL processes, data preprocessing, data visualization with Tableau
- **Version Control & Collaboration:** Git, GitHub, GitLab

SOFT SKILLS

- Teamwork
- Problem-solving
- Adaptability
- Time management
- Presentation skills
- Work under pressure
- Responsibility & accountability

LANGUAGE

- **English level:** Intermediate –capable of reading technical materials, following conversations, and using domainspecific vocabulary in simple discussions.

SOME ACTIVITIES

- **University Entrance Exam Support Volunteer (2023 & 2024)**
  - Provided guidance and support to high school students during university entrance examination period
  - Assisted with exam venue setup, student registration, and information distribution
  - Helped maintain order and provided emotional support to nervous examinees
- **Blood Donation Volunteer (2024)**
  - Regular blood donor with 3 successful donations (350ml each)
  - Participated in university blood drive events organized by Red Cross Vietnam
  - Helped promote blood donation awareness among students and community
- **Environmental Conservation Volunteer (2021 & 2022)**
  - Participated in monthly 'Green Saturday' campus cleanup activities
  - Educated local community about plastic waste reduction and recycling

HOBBIES AND INTERESTS

- Playing the guitar
- Photography using DSLR camera
- Playing basketball